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655.662 — Introduction to Healthcare Systems Engineering
Module 6 | M6A1: Problem Set — Kossiakoff Problem 13.2

Advanced development is the process of developing a product or technology beyond a conceptual stage, yet prior to establishing a final design. Advanced development's purpose is not to develop the product itself; however, it does remove the uncertainty that could surface later, when correcting errors may result in significant expense.

Six separate areas of focus provide advanced development with its other name, risk reduction. As stated by the class materials, the goal of prototyping and competition is to lower technical, cost and schedule risks. Each area listed below has been associated with both a type of risk and an example of how each of those types of risk relates to the PD-RPM (peritoneal dialysis remote patient monitoring) system, developed throughout this course.

1. Identify and rank high-risk items. The first part of advanced development is determining where the risk truly is, so that instead of distributing the resources equally throughout the system, effort can be focused on those few aspects that require attention. By doing so, programmatic risk, the risk of using advanced development funds on the incorrect elements, is reduced. An example of this is the technology readiness survey that was conducted for PD-RPM's 20 physical elements. While 19 of the physical elements had a technology readiness level (TRL) rating of 9 (commercial cuffs, scales, workstations, etc.), only one of the 20 had a TRL rating of 6. That element was the HG-1 home gateway, with its cellular store-and-forward firmware. In addition, the companion risk associated with this element, "silent loss" of home data delivery, had a rating of $4 \times 4 = 16$. Therefore, advanced development was essentially confined to the HG-1 home gateway, and the remaining commercial elements were allowed to proceed without additional development.

2. Requirements analysis and flow-down to a validated specification. System requirements are determined through analyzing operational requirements. These system requirements are then tested to ensure that they exist as intended and are tied directly to their respective sub-functions. Verification risk is reduced as ambiguities are identified on paper versus at the time of acceptance testing. An example of this is the PD-RPM operational requirement that requires each completed treatment record be transferred to the monitoring center within 30 minutes of completion. Flow down to this requirement resulted in four different timed functions including a pairing timeout of 10 seconds, a buffer write confirm period of 2 seconds, a 5-minute retry interval over 12 attempts for uploading treatment records, and an ingest delay until available of 90 seconds. However, none of the requirements made provisions for carrier re-registration delays (4 minutes). That gap only became visible once the budget was written out.

3. Component trade studies. Engineering trade studies are the primary tool employed in advanced development. Alternatives to various component selections are analyzed and compared based upon weighted criteria that relate directly to functional requirements. Cost and schedule risks are lowered as alternatives that lose are removed prior to purchasing any goods or supplies. An example of this is the PD-RPM connectivity trade study that evaluated four alternative methods of connecting patient homes to the monitoring center using four different criteria. The method employing a tethered smartphone connection was less costly than all other options by a factor of four; however, it was eliminated from consideration due to its inability to achieve a required KPP (data completeness) of 95 percent (91 percent achieved). If the same shortfall had been discovered in the field after 150 gateways were deployed, eliminating the tethered option would have resulted in a recall. Eliminating it on paper required one week of analysis.

4. Modeling and simulation. Measures related to feasibility and performance are quantified before actual hardware exists. Technical risk is thereby decreased at minimal cost and what was previously considered a design argument becomes a numeric value. An example of this is a discrete event simulation model used to quantify the buffer sizing necessary for a PD-RPM fleet of 150 patients. The model utilized data collected from previous sessions and data collected from measuring actual carrier outages. The results indicated that if a 72-hour buffer existed in the original concept, data loss would occur in approximately 8 percent of simulated months; whereas, if a 14-day buffer existed in the concept, no losses would occur. Prior to building any enclosures, the buffer size requirement was modified.

5. Prototype development for critical components. Explorative prototypes are built in order to allow users to explore requirements for products, while experimental prototypes are built to determine whether products will operate as planned. Both types of prototypes decrease technical risk and the experimental prototype can serve as a substitute for a formal requirements document. An example of this is an experimental HG-1 that represented a single board gateway operating with production firmware and no housing enclosure. The experimental HG-1 operated against 20 simulated cyclers on a carrier test network for 6 weeks under controlled conditions involving intentional SIM deactivation and intentional firmware failure. Results indicated that store-and-forward operations continued even though a 9-day outage occurred, and liveness heartbeat detected a dead link in less than 35 minutes.

6. Test planning for development tests and integration build-up planning. A test and evaluation plan along with an integration build up plan reduces integration risk, the risk that the first time that elements meet is also the first time someone discovers they don’t fit together properly. An example of this is the PD-RPM integration build up plan. The plan builds up from HG-1 alone on the bench; HG-1 with the cycler data port; HG-1 with the Bluetooth peripherals; HG-1 end-to-end to ingest service over a live carrier; monitoring center services against simulated feeds; full chain with simulated patients; full chain with clinicians in loop; and finally limited clinical evaluations with real patients last. With each build up step adding one new element, defects discovered can be attributed to the item recently added.

	Activity	Risk class retired	What it produced for PD-RPM
1	Identify and rank high-risk elements	Programmatic	HG-1 isolated at TRL 6 as the one element needing the phase
2	Requirements flow-down	Verification	30-minute transfer budget shown to omit carrier re-registration
3	Trade studies	Cost and schedule	Cheapest alternative disqualified on a KPP before procurement
4	Modeling and simulation	Technical	Buffer resized from 72 hours to 14 days before tooling
5	Prototype development	Technical	Store-and-forward verified across a nine-day outage
6	Development test planning	Integration	Eight-step build-up isolating each defect to one element

The common thread is that each activity replaces a belief with a measurement while the cost of being wrong is still small. The phase is called risk reduction precisely because nothing in it is meant to ship: the prototype has no enclosure, the simulation has no patients, the build-up stops short of production. What it delivers is a design the program can commit to and a record of the uncertainties that no longer have to be carried forward.